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Approximating discrete probability distributions with dependence trees

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Abstract

A method is presented to approximate optimally an n-dimensional discrete probability distribution by a product of second-order distributions, or the distribution of the first-order tree dependence. The problem is to find an optimum set of n - 1 first order dependence relationship among the n variables. It is shown that the procedure derived in this paper yields an approximation of a minimum difference in information. It is further shown that when this procedure is applied to empirical observations from an unknown distribution of tree dependence, the procedure is the maximum-likelihood estimate of the distribution.

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